

Shikha Nukala

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Education

University of Michigan, College of Engineering

Bachelor of Engineering in Computer Science

Ann Arbor, MI

May 2026

- Awards/Honors: Four terms Dean's List College of Engineering, University Honors
- Courses Completed: Foundations of Computer Science, Cryptography, Computer Security, Compilers, Computer Networks, Quantum Computing, Data Structures and Algorithms, Discrete Mathematics, Computer Organization

Relevant Experience

Delta Dental of Michigan, Indiana, and Ohio

Application Development Intern

Okemos, MI

May 2024-Aug. 2024

- Converted and modified existing SOAP APIs to REST for Delta Dental's Dental Provider Locator website and used Java and Postman to implement code and run tests to the provider database.
- Wrote an Angular application using Javascript, HTML, and CSS to redesign the Provider Locator website using UI/UX engineer's design and wrote smoke tests to test functionality.
- Integrated REST APIs to display database results from dental provider searches in the new Angular application.
- Collaborated with App Dev and product teams to perform API testing using Postman with secure token handling and error logging for threat detection.

MedLaunch

Project Co-Lead

Ann Arbor, MI

Sept. 2023-present

- Designed pages of an application in Figma for tracking and improving the mental health and wellness of students at the University of Michigan partnering with a Wolverine Wellness coach and UM medical student.
- Implemented Figma designs and additional requests from community partners in React and Expo Go to create an Android and iOS friendly application.
- Gathered feedback from partners and provided monthly updates and trials of the application.

Roosevelt Innovations

Knowledge Science Intern

Okemos, MI

May 2023-Nov. 2023

- Wrote a Rules engine and corresponding grammar to execute within a custom-built IDE and provide more functional and user-friendly business rules syntax.
- Used Drools and ANTLR to pilot rule-based systems to reduce open-source documentation usage and create a completely Roosevelt-developed claims-processing tool.
- Modified a custom Angular IDE to guard against XSS through DOM sanitization, secure component rendering, and proper encoding of user-defined rules.

Projects

Network Defense & Threat Detection:

- Deployed Docker environments and Python sockets to simulate secure server-client communications and identify vulnerabilities using a VM.
- Configured IDS/IPS rules to alert on malicious traffic patterns such as DoS attempts.

Application Security & Reverse Engineering:

- Developed PowerShell and Bash scripts for log monitoring and detection of privilege escalation attempts.
- Used DEP with preset data for objects and removed extra memory allocation to prevent ROP attacks.
- Investigated malware samples and learned how to extract IoCs with reverse engineering.

Software Engineering Projects:

- Created a web application to match homeless shelters with restaurants who have surplus food using React, Firebase, Google APIs, and developed a matching algorithm to match users based on profile information and user criteria.
- Implemented complex quantum bit circuits using the Python qiskit library to model breaking RSA encryption and solving search problems in exponentially faster time.

Technical Languages: Java, Javascript, C++, Python, HTML, CSS, R, SQL, MATLAB, ANTLR

Leadership Experience

M-HEAL Team DRIP

Technical Team Lead

Ann Arbor, MI

Sept. 2024-present

- Leading website development for a hospital in Puerto Penasco, Mexico to raise awareness for diabetic amputees.
- Overseeing design and 3D printing of a prosthetic knee for diabetic amputees that is low-cost and highly durable.

Sahana Classical Dance and Music

Logistics Committee Lead

Ann Arbor, MI

Sept. 2022-Apr. 2024

- Arranged three hour dance competition shows for all Indian artistic groups at the University of Michigan.
- Booked and decorated Michigan theater, fundraised, coordinated performers, and sold 1000+ tickets.